Year 10 Foundation

REVISION PACK

Sets 1

Name: _____

Class: _____

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2)

Worked example: completing a Venn diagram

$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$ $A = \{1, 2, 3, 4\}$ $B = \{2, 4, 6, 8\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

(a)	**Step 1**	In both: the middle	$2, 4$
	Step 2	Only A: left	$1, 3$
	Step 3	Only B: right	$6, 8$
	Step 4	The rest: outside	$5, 7$
(b)	**Step 1**	Answer: in both	$\{2, 4\}$
(c)	**Step 1**	Answer: in A or B	$\{1, 2, 3, 4, 6, 8\}$
(d)	**Step 1**	Not in A	$5, 6, 7, 8$
	Step 2	Answer	4

Practice: completing a Venn diagram

1.1 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ $A = \{2, 4, 6, 8, 10\}$ $B = \{1, 2, 3, 4, 5\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

1.2 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ $A = \{1, 3, 5, 7, 9, 11\}$ $B = \{3, 6, 9, 12\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

1.3 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ $A = \{1, 4, 9\}$ $B = \{1, 2, 3, 4, 5, 6\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

1.4 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ $A = \{4, 8, 12\}$ $B = \{2, 3, 4, 5, 6\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

1.5 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$ $A = \{5, 10, 15\}$ $B = \{1, 3, 5, 7, 9\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

1.6 $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ $A = \{3, 6, 9\}$ $B = \{2, 3, 5, 7\}$

(a) Complete the Venn diagram. (b) List the members of $A \cap B$. (c) List the members of $A \cup B$. (d) Find $n(A')$.

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2)

Worked example: problems with two sets

25 students were asked if they play football (F) or tennis (T). 14 play football. 10 play tennis. 4 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

(a)	**Step 1**	Both: the middle	4
	Step 2	Football only	$14 - 4 = 10$
	Step 3	Tennis only	$10 - 4 = 6$
	Step 4	Neither: outside	$25 - (10 + 4 + 6) = 5$
(b)	**Step 1**	Add the two "only" numbers	$10 + 6 = 16$
	Step 2	Answer	16

Practice: problems with two sets

1.7 30 students were asked if they play football (F) or tennis (T). 15 play football. 11 play tennis. 4 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

1.8 40 students were asked if they play swimming (S) or running (R). 22 play swimming. 15 play running. 7 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

1.9 28 students were asked if they play chess (C) or drama (D). 13 play chess. 11 play drama. 3 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

1.10 35 students were asked if they play netball (N) or hockey (H). 20 play netball. 12 play hockey. 6 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

1.11 25 students were asked if they play volleyball (V) or golf (G). 14 play volleyball. 9 play golf. 5 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

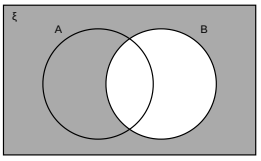
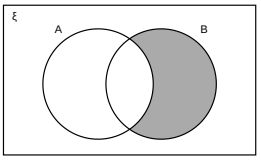
1.12 50 students were asked if they play basketball (B) or karate (K). 26 play basketball. 20 play karate. 8 play both.

(a) Complete the Venn diagram. (b) How many students play only one of the sports?

Sets 1 · Set notation and Venn diagrams (Lessons 9.1, 9.2)

Worked example: shading regions

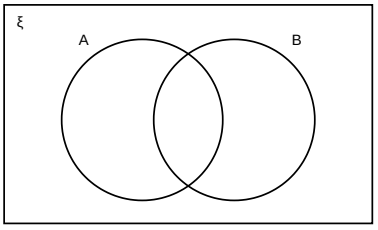
Shade the region on each Venn diagram. (a) B' (b) $A' \cap B$

(a)	**Step 1** B': outside B	
	Step 2 Shade everything not in B	
(b)	**Step 1** A': outside A	
	Step 2 $\cap B$: and inside B	
	Step 3 Shade B only	

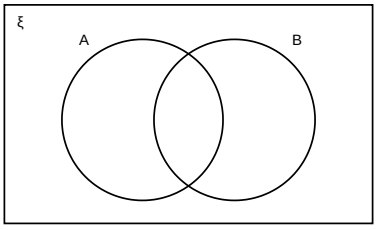
Practice: shading regions

1.13 Shade the region on each Venn diagram.

(a) A'

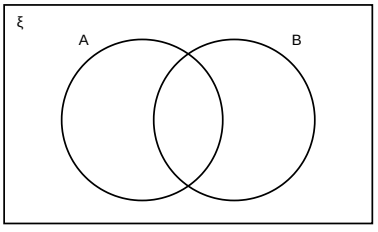


(b) $A \cap B$

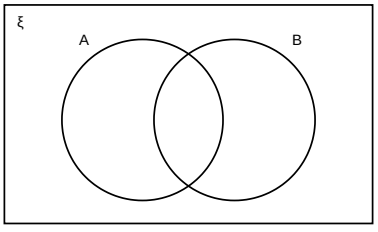


1.14 Shade the region on each Venn diagram.

(a) $A \cup B$

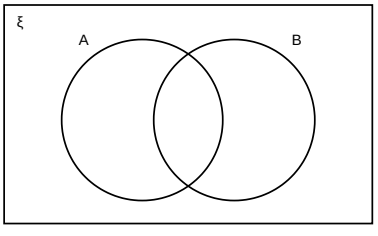


(b) B'

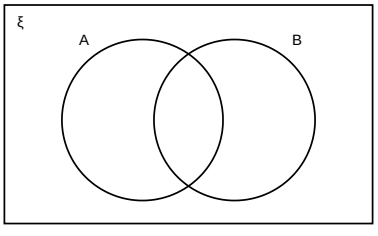


1.15 Shade the region on each Venn diagram.

(a) $A \cap B'$



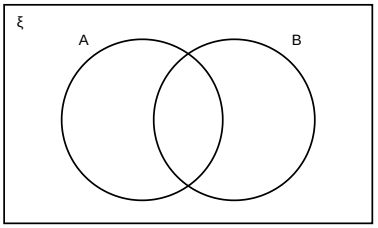
(b) A'



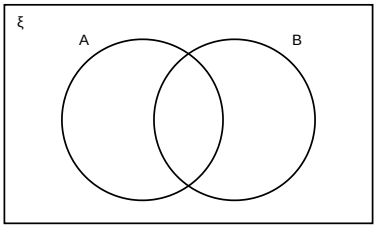
Practice: shading regions (continued)

1.16 Shade the region on each Venn diagram.

(a) B'

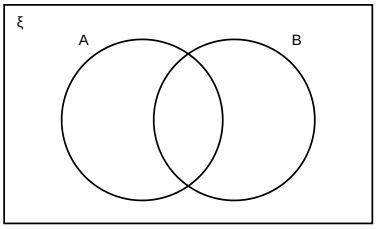


(b) $(A \cup B)'$

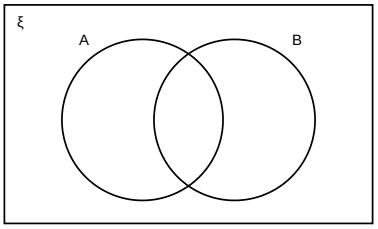


1.17 Shade the region on each Venn diagram.

(a) $A \cap B$

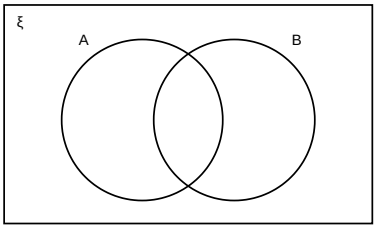


(b) $A' \cap B$



1.18 Shade the region on each Venn diagram.

(a) $A \cup B$



(b) $A \cap B'$

